KM PRIYANKA

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Professional Summary

Results-driven VLSI Engineer with hands-on experience in digital IC design, physical and logical synthesis, and hardware optimization. Skilled in managing ASIC design flows, timing analysis, LEC, and PPA optimization. Proficient in industry-standard EDA tools including Cadence Genus, Innovus, Virtuoso, Conformal, and Joules. Adept at resolving timing, congestion, and scan chain issues across multiple project stages. Seeking to contribute technical expertise to cutting-edge semiconductor design and development.

Work Experience

Skychip Silicon Pvt. Ltd
Physical Design Engineer — *April 2025 – June 2025*

Project: **Advanced 7nm SoC Integration and Timing Closure**

Description: This project involved the development of a high-performance SoC using the cutting-edge with 7nm technology node. The design encompassed 85 macros, comprising over 1 million instances spread across 11 metal layers, and featured two distinct clock domains operating at 1.2GHz and 560MH, emphasizing the need for precise timing and robust design integrity.

Role and Responsibilities:

- **Low-Power Synthesis and Signoff Ownership:** Led CLP and physical synthesis for low-power sub-systems, achieving PPA goals; managed LEC/vCLP closure from RTL to final netlist and delivered timing-closed handoff to Innovus.
- **System Integration and Cross-Team Collaboration:** Integrated 35+ partitions and brought up Flat vCLP session; collaborated with RTL, SoC PD, and implementation teams to resolve UPF issues and drive full-chip signoff.

Challenges:

- **Design Verification Challenges:** Navigated complexities in Synthesis, Static Timing Analysis (STA), Logical Equivalence Checking(LEC), and CLP.
- **Timing Closure Congestion Resolution:** Addressed and optimized timing issues while resolving congestion challenges for improved performance.

Abhaytam Smartworks Pvt. Ltd.
Physical Design Engineer — *June 2024 – March 2025*

Project: **TSPC Flip-Flop Based Design and Integration using Cadence Tools**

Description: Designed a True Single-Phase Clock (TSPC) flip-flop in Virtuoso targeting low power and high speed. Integrated it into the RTL, replacing standard DFFs, and verified through full digital flow.

Role and Responsibilities:

- Created and characterized the TSPC flip-flop at the transistor level. Replaced flip-flops in RTL, synthesized in Genus.

Challenges:

- Faced glitches due to insufficient transistor sizing and fixed them via iterative tuning. Handled missing constraints, DRV violations, and tool integration errors during timing closure.

Projects

1. Studies on Aging Effects of ZnON-Based Thin-Film Transistors — **Jan. 2023 – Jul. 2024**

Researched and validated parameter extraction in MATLAB, analyzed aging effects under various conditions, extracted key parameters (Vth, mobility, Ion current), and used curve fitting to model deviations for accurate long-term transistor behavior simulation, ensuring optimal circuit design.

2. Design and Analysis for Dual Rail Technology for Security Purposes **Apr. 2023 – May. 2023**
Enhanced hardware security through dual rail technology and master-slave circuitry while improving power efficiency with clock-gating.

Technical Skills

Skills

- Physical Design (ASIC and PnR Flow, Congestion resolving)
- Static Timing Analysis (STA)
- Synthesis (Logical and Physical)
- DFT Stitching
- Clock gating, CLP Flow and validation, FV and LEC flow and checks.
- Verilog HDL
- Digital IC Design
- Digital Electronics

Tools Used

- Cadence (Genus, Virtuoso, Innovus and, PrimeTime)
- Low Power Conformal
- EDA
- Python
- MATLAB
- Icarus Verilog
- Xilinx Vivado

Education

National Institute of Technology, Calicut
Master of Technology in Electronics Design and Technology *Aug. 2022 – Jul. 2024*

Dr. A.P.J. Abdul Kalam Technical University, Lucknow
Bachelor of Engineering in Electronics and Communication *Jul. 2016 – Sept. 2020*

Publications

Advances in Smart Sensor, Signal Processing and Communication Technology. Accepted for publication by AIP Publishing.